



Module #3 Investment replica

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Portfolio replication in short

We want to track a given portfolio («target portfolio») without knowing its holdings

We do not know either which securities the portfolio contains or what weight each security has

We can only observe reported returns: for us, the target portfolio is a complete black box

We want to crack this black box and see how it is made inside



Why on earth do
that???

INSTITUTIONAL INVESTOR

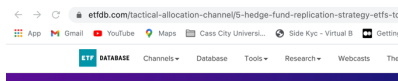


Andrew Beer

Why Managed Futures Funds Are Ripe for Replication

Research shows that replication, rather than investments in sin manager hedge funds, may make sense for most allocators, wri Beer.

September 5, 2023



BEYOND BASIC BETA CHANNEL 5 Hedge Fund Replication Strategy ETFs to Consider

Ben Hernandez Apr 18, 2019

Most might consider 13 to be an unlucky number, but for hedge funds, it mark best first quarter its had in as many years. According to the Hedge Fund Rese index, which tracks the performance of hedge funds, it rose 5.7 percent in Q1 its highest since 2006.



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The Rise of Hedge Fund ETFs – Democratizing Alternative Strategies



Aous Labbane

Executive | Founder | Driving Innovation in Derivatives, Investment Strategies and ETFs



January 27, 2026

stitutional Investor

Portf

PORTFOLIO



Julie Segal

Hedge Fund Replication Worked in January. Here's W

Replication passed a significant test in the volatile month of Jar investors sent the stock of GameStop soaring and some equity l reeling.

February 12, 2021

om/2021/01/replicate-hedge-funds/

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MiB: How to Replicate Hedge Funds in an ETF

January 26, 2021 10:00am by Barry Ritholtz

Podcasts



22 GENNAIO - 1 ORA 14 MIN

Andrew Beer on the Hedge Fund Industry (Podcast)
Masters in Business

Riproduci

Vedi altro

Motivation 1

– Alternative Investments Clones

YOU WANT TO BE EXPOSED TO THE SAME KIND OF FINANCIAL RISK AND RETURN WHILE AVOIDING HIGH FEES AND LIQUIDITY RESTRICTIONS TYPICAL OF ALTERNATIVE INVESTMENTS (EG HEDGE FUNDS)

YOU HAVE A LONG POSITION IN AN ALTERNATIVE INVESTMENT FUND, OR IN A PORTFOLIO OF ALTERNATIVES (FUND OF FUNDS), AND YOU MAY WISH TO HEDGE THE RISK OF A DOWNTURN

FOR RISK ANALYSIS PURPOSES YOU WANT TO UNDERSTAND WHICH RISK FACTORS DRIVE THE ALTERNATIVE FUND

Motivation 2 – Index Tracking

- consequence
of Motivation 1

You may want to track a very illiquid index - which is very costly if not impossible from the execution point of view

You may want to track a very big index - which means having huge Assets under Management (AuM): many funds have less than 100 million euros/US dollars, and you typically need billions, in order to physically replicate an index

iMGP DBi Managed Futures Strategy ETF DBMF ★★★★★

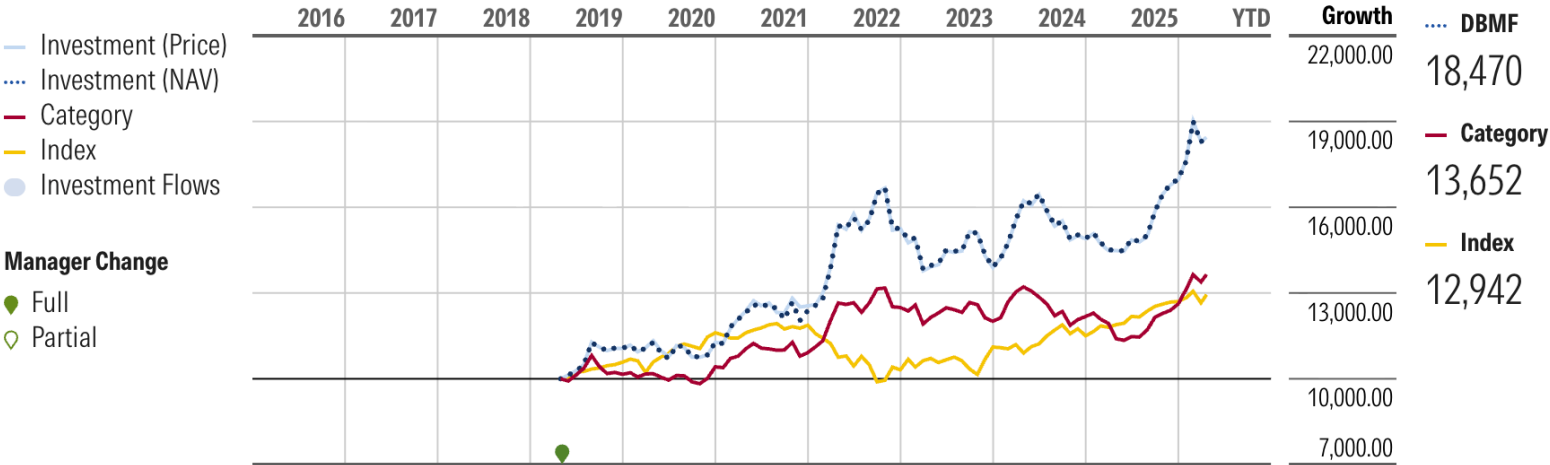
Performance | Rating as of Mar 31, 2026 | [See iM Global Investment Hub](#) >

[Quote](#) [Chart](#) [Fund Analysis](#) [Performance](#) [Sustainability](#) [Risk](#) [Price](#) [Portfolio](#) [People](#) [Parent](#)

Performance

Returns Distributions

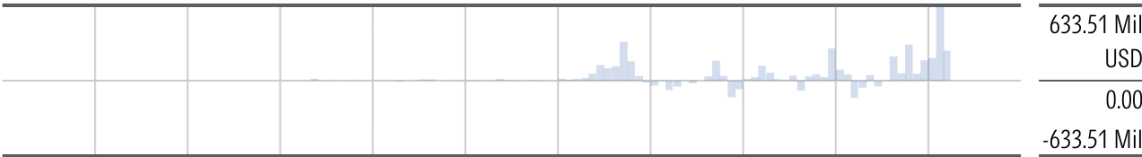
Growth of 10,000



Manager Change

- Full
- Partial

Investment Flows



ProShares Hedge Replication ETF HDG ★★★

Performance | Rating as of Mar 31, 2026 | [See ProFunds/ProShares Investment Hub](#) >

[Quote](#) [Chart](#) [Fund Analysis](#) [Performance](#) [Sustainability](#) [Risk](#) [Price](#) [Portfolio](#) [People](#) [Parent](#)

Performance

Returns Distributions

Growth of 10,000

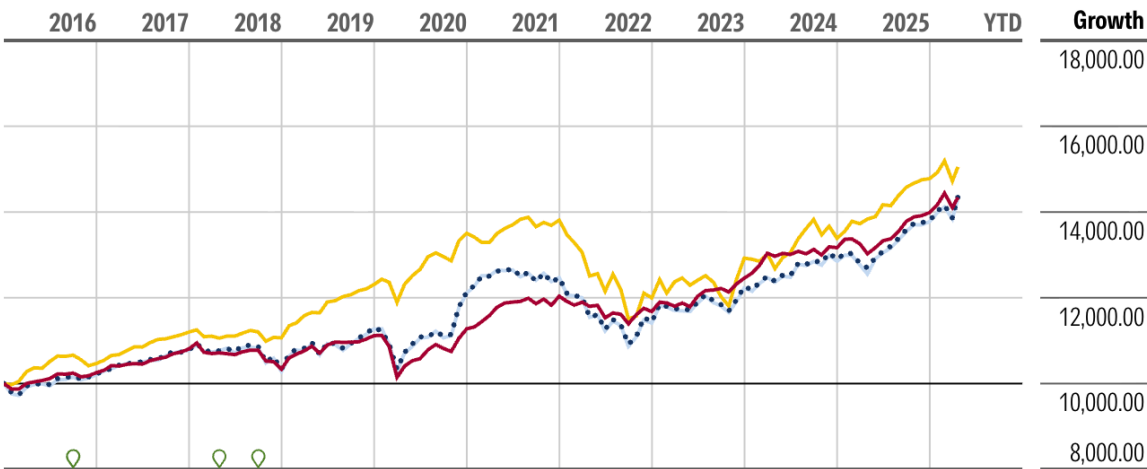
Cass City University of London



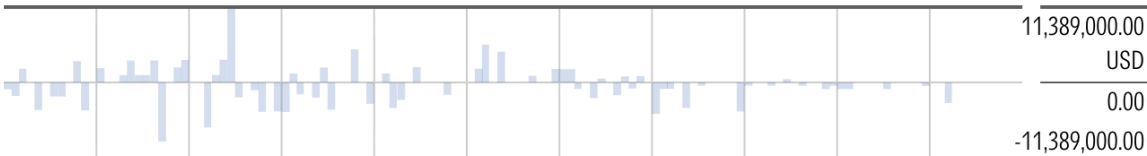
- Investment (Price)
- Investment (NAV)
- Category
- Index
- Investment Flows

Manager Change

- Full
- Partial



Investment Flows



ALTERNATIVE · ETF

QAI

NYLI Hedge Multi-Strategy Tracker ETF

NAV

\$35.48

As of 04/22/2026

1 Day Change

▲ **0.17** 0.49%

YTD

▲ **5.85%**

	QTR	YTD	1Y	3Y	5Y	10Y	SI
At NAV	1.77	1.77	10.52	8.00	3.49	3.30	3.26
At Market Price	1.82	1.82	10.54	8.02	3.47	3.31	3.26
NYLI Hedge Multi-Strategy Index	1.99	1.99	11.55	8.85	4.22	4.05	4.08

Motivation 3 – UCITS clones

Say that the company has an alternative fund for institutional investors, which invests in instruments that are not UCITS, that is, not eligible by the MIFID regulation (e.g. private debt, other alternative funds, etc.)

The company wants to create a retail products, that is, a UCITS vehicle (ie mutual fund) with similar financial characteristics, but made up of liquid instruments, e.g. Futures, ETFs, etc.,

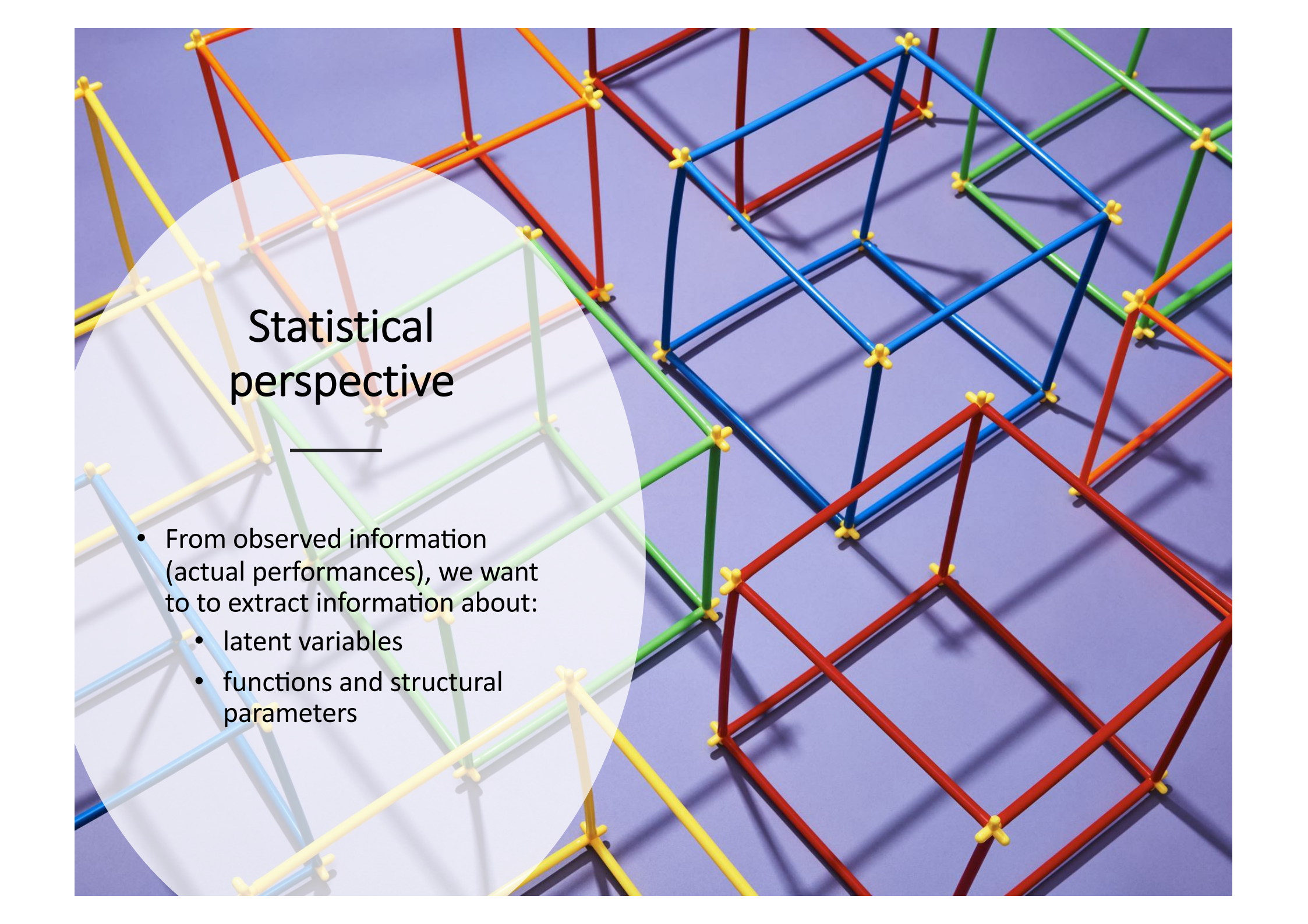
Motivation 4 – Risk Management

You want a standard tool that finds the key liquid financial factors behind any investment, even if the underlying assets are not disclosed

It can be used as an alternative to standard factor models

Useful during due diligences, or litigations

Useful for simple scenario simulation (instead of using thousands of securities, you simulate just a few factors)



Statistical perspective

- From observed information (actual performances), we want to extract information about:
 - latent variables
 - functions and structural parameters



Linear clones

- Portfolio return is a linear combination of asset returns
- So we need a linear model, a linear clone
- A linear clone is constructed by.
 - estimating some regression model using a time series of prior index and factor returns
 - using the resulting coefficient estimates as portfolio weights (ie investment positions in the underlying factors)
- Although index tracking is driven from the financial industry, it is in fact a pure signal processing problem: a regression of the (historical) data subject to some portfolio constraints with some caveats and particularities
- «**Sparse portfolio tracking**»: we use a small number of assets to (approximately) replicate a portfolio or an index

Using Futures, in short

- You have 100 millions EUR (normal mutual fund)
- Buy Futures contracts exposing yourself to 100 millions of S&P 500: only the money corresponding to the margins is paid
- Futures margin are typically 3-12% of the notional value of the contract
- The rest is invested in short-term "collateral" bonds that hopefully yield something and can easily sold in case of margin call
- So your financial exposure is (say): 90 short term bonds + 100 S&P 500 + 10 cash (margins) = 200% exposure
- Using Futures, you can financially expose yourself far beyond your capital: welcome to the LEVERAGE world

You can do
that
according to
UCITS/MIFID

Max leverage =

- 200%

Or

- Max 1M VaR(99%)
= 20%





Let's look at
the data

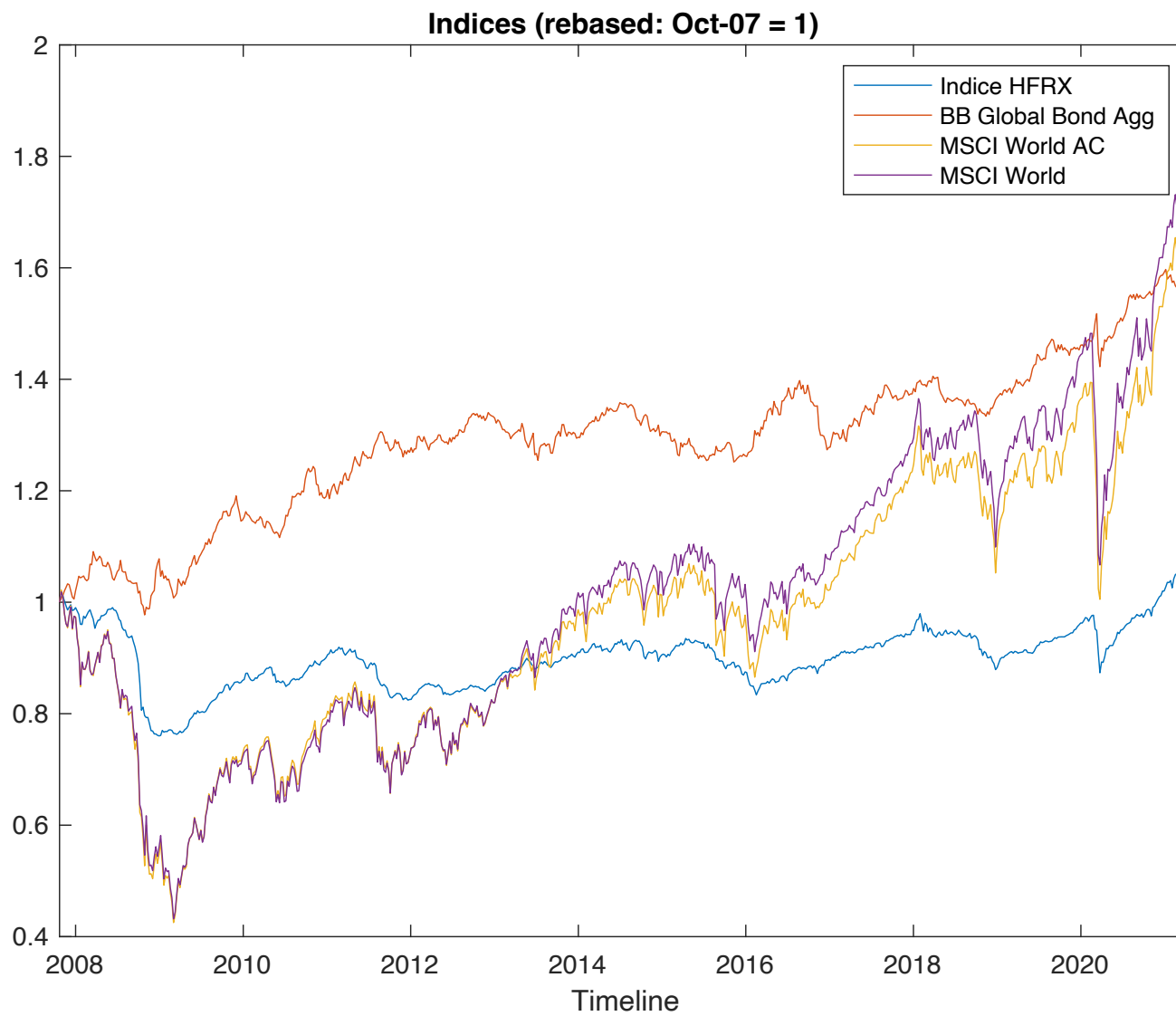


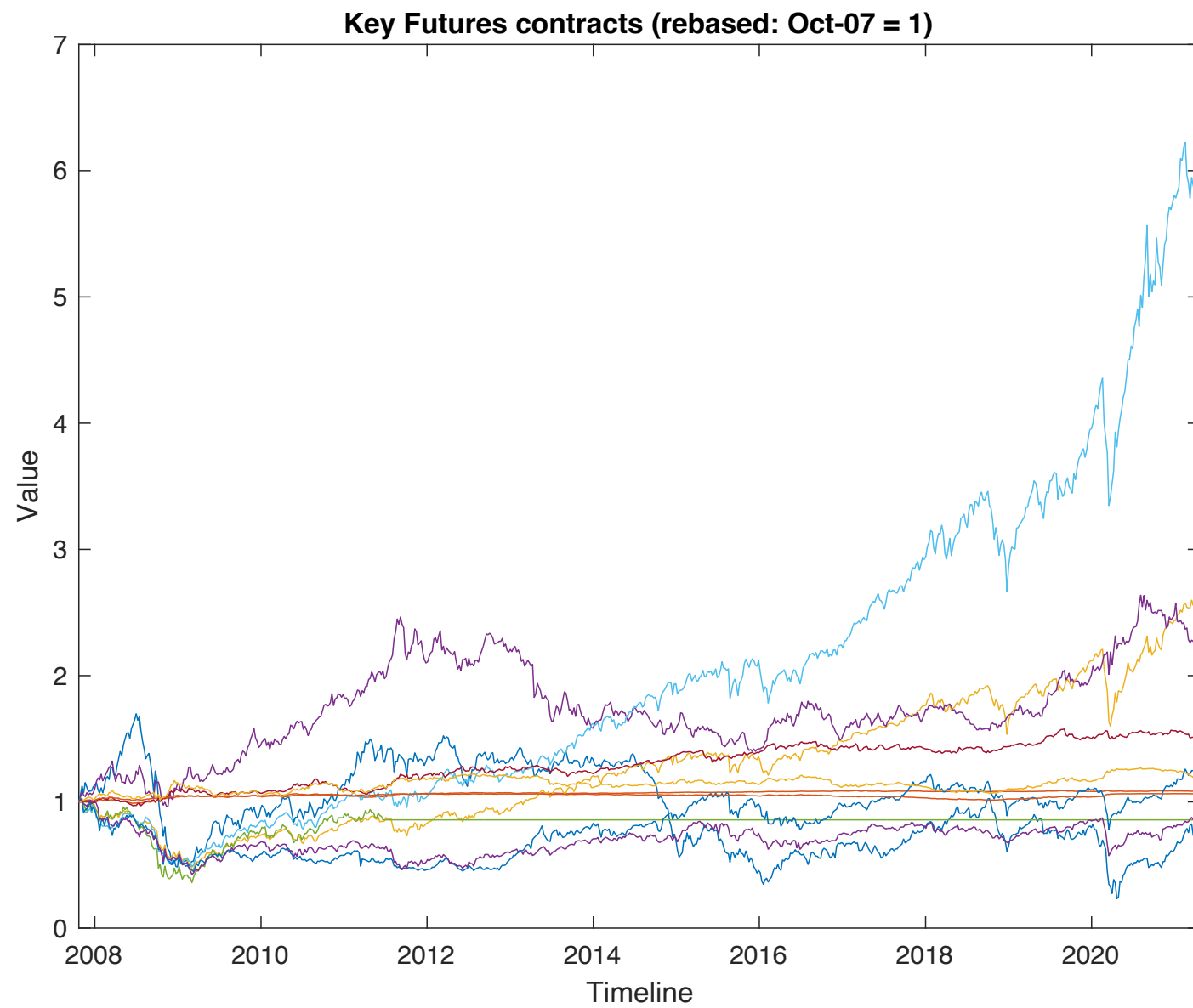
Data overview

Weekly data from Bloomberg

(Oct-2007/Apr-2021)

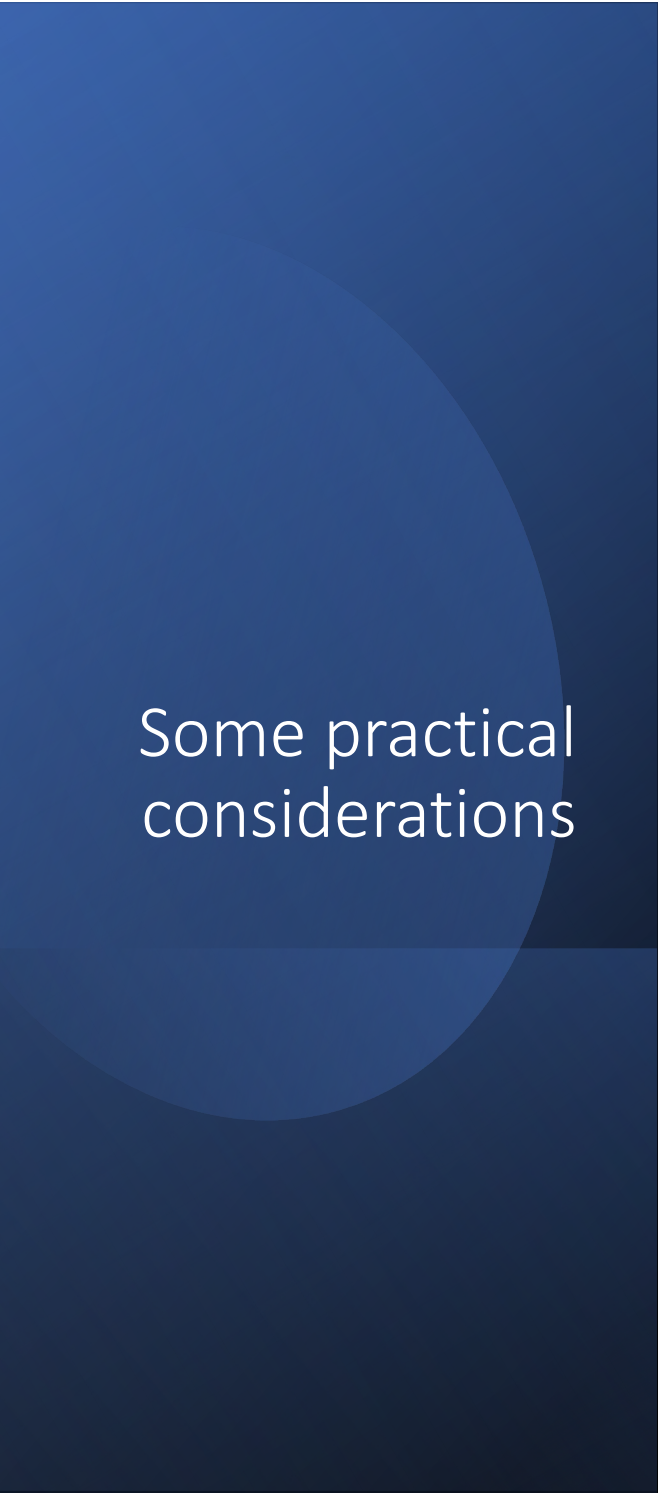
- A global Hedge Fund Index
- Global indices (bond/equity)
- Key Futures contracts







Coding session starts |



Some practical considerations

- We focus on the algorithm (after all, this is a Machine Learning lab for Fintech - we cannot go much farther in the territory of portfolio construction and trading)

However

- You can do a better job if you go deeper with regard to:
 - Trading and transaction costs/revenues
 - Rebalancing times and methods
 - Tricks for minimizing execution costs

Trading costs

- Futures contracts are the most liquid and least expensive financial instruments to trade, on average
- Assuming that we are a mutual fund (or any other «UCI», Undertakings in Collective Investment product) of decent size – say >50 millions EUR – using automated program trading, direct trading costs can be 2-4 bps = 0.02%-0.04%





Rebalancing

- It can be weekly, monthly, quarterly, or event –based
- The change in the composition of the holding portfolio during rebalancing can be controlled by including a turnover penalty
 - we already take this into account if we use Machine Learning techniques with Regularization, such as LASSO (👉 shrinkage of the weights towards 0)
 - eliminate very small trades and eventually normalize the weights (preserve proportions among assets)
 - avoid trading if the distance between the vector of the "old" weights and the new ones is less than a certain threshold

Rollover effects

- To maintain a constant exposure to the Futures contract, an expiring contract needs to be rolled into a new contract
- You face rollover effects when you switch from the contract that is close to expiration to another contract in a further-out month
- It can be positive or negative
- When you close the contract that expires soon, it is priced close to Spot, but the new contract that you enter into may be priced above or below Spot
- It has to do with contango and backwardation:
 - Contango = the forward price of a contract is higher than the spot price
 - Backwardation = the spot price is higher than future price



Some useful references

- Lan Wu, Yuehan Yang, Hanzhong Liu, (2014) «Nonnegative-lasso and application in index tracking», Computational Statistics & Data Analysis, Volume 70, <https://doi.org/10.1016/j.csda.2013.08.012>
 - Tibshirani, R. (1996). Regression shrinkage and selection via the lasso J. R. Statist. Soc. B (2011) 73, Part 3, pp. 273–282 <https://tibshirani.su.domains/ftp/lasso-retro.pdf>
 - Ali N. Akansu, Sanjeev R. Kulkarni, Dmitry M. Malioutov, (2016), «Financial Signal Processing and Machine Learning», Wiley-IEEE Press
 - Roncalli, Thierry and Weisang, Guillaume, (2009) «Tracking Problems, Hedge Fund Replication and Alternative Beta». Available at SSRN: <https://ssrn.com/abstract=1325190> or <http://dx.doi.org/10.2139/ssrn.1325190>
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